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(i) Using monthly data from February 1978 through December 1988 to estimate the following model:

$$\log(\text{chnimp}) = \delta_0 + \delta_1 \log(\text{chempi}) + \delta_2 \log(\text{gas}) + \delta_3 \log(\text{rtwex}) + \delta_4 \text{befile6} + \delta_5 \text{affile6} + \delta_6 \text{afdec6} + u$$

```
. reg lchnimp lchempi lgas lrtwex befile6 affile6 afdec6
```

Source	SS	df	MS	Number of obs	=	131
				F(6, 124)	=	9.06
Model	19.4051607	6	3.23419346	Prob > F	=	0.0000
Residual	44.2470875	124	.356831351	R-squared	=	0.3049
				Adj R-squared	=	0.2712
Total	63.6522483	130	.489632679	Root MSE	=	.59735

	lchnimp	Coefficient	Std. err.	t	P> t	[95% conf. interval]
	lchempi	3.117193	.4792021	6.50	0.000	2.168718 4.065668
	lgas	.1963504	.9066172	0.22	0.829	-1.598099 1.9908
	lrtwex	.9830183	.4001537	2.46	0.015	.1910022 1.775034
	befile6	.0595739	.2609699	0.23	0.820	-.4569585 .5761064
	affile6	-.0324064	.2642973	-0.12	0.903	-.5555249 .490712
	afdec6	-.565245	.2858352	-1.98	0.050	-1.130993 .0005028
	_cons	-17.803	21.04537	-0.85	0.399	-59.45769 23.85169

Econometric Interpretation

1. Demand-Side Controls

- **Chemical production index (*chempi*)** is highly significant and economically meaningful. A 1% increase in chemical production is associated with a 3.12% increase in Chinese barium chloride imports, holding other factors constant. This reflects strong industrial demand elasticity.
- **Gasoline production (*gas*)** is not statistically significant, suggesting it does not explain variation in barium chloride imports in this context.
- **Exchange rate index (*rtwex*)** is positively significant. A stronger dollar (higher *rtwex*) is associated with increased imports, consistent with economic theory: a stronger domestic currency makes foreign goods cheaper.

2. Policy Dummies

- *befile6* and *affile6* are statistically insignificant, implying that imports were not unusually high before the filing, nor did they change immediately after.
- *afdec6* is marginally significant at the 5% level, with a negative coefficient. This suggests that imports declined by approximately 56.5% in the six months following the ITC's favorable decision for the U.S. industry. This supports the hypothesis that the ruling had a deterrent effect on Chinese imports.

3. Model Fit

- **R-squared** = **0.305**, indicating that about 30.5% of the variation in log imports is explained by the model. While not high, this is reasonable for monthly trade data with policy shocks.
- **F-statistic** = **9.06**, with **p** < **0.001**, confirming that the model is jointly significant.

The regression provides strong evidence that:

- Chemical production and exchange rate movements significantly influence Chinese imports.
- The ITC's decision in favor of the U.S. industry had a statistically and economically meaningful deterrent effect on imports.
- There is no evidence of strategic import surges before filing or immediate changes post-filing.

(ii) Add a linear time trend to the above equation. Are any variables, other than the trend, statistically significant?

```
. reg lchnimp lchempi lgas lrtwex befile6 affile6 afdec6 t
```

Source	SS	df	MS	Number of obs	=	131
-----+-----				F(7, 123)	=	9.95
Model	23.0142898	7	3.28775569	Prob > F	=	0.0000
Residual	40.6379584	123	.330389906	R-squared	=	0.3616
-----+-----				Adj R-squared	=	0.3252
Total	63.6522483	130	.489632679	Root MSE	=	.5748

lchnimp	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
-----+-----						
lchempi	-.6862364	1.239711	-0.55	0.581	-3.140169	1.767696
lgas	.4656786	.8761779	0.53	0.596	-1.268662	2.200019

lrtwex		.0782237	.47244	0.17	0.869	-.8569423	1.01339
befile6		.09047	.2512887	0.36	0.719	-.4069406	.5878805
affile6		.0970062	.2573131	0.38	0.707	-.4123294	.6063417
afdec6		-.3515018	.2825417	-1.24	0.216	-.9107758	.2077722
t		.0127058	.0038443	3.31	0.001	.0050963	.0203153
_cons		-2.367526	20.78216	-0.11	0.909	-43.50455	38.7695

Econometric Interpretation

1. Time Trend Dominance

- The time trend coefficient is positive and highly significant ($p = 0.001$), indicating a systematic upward drift in Chinese barium chloride imports over time, independent of other explanatory variables.
- This suggests that imports were rising steadily across the decade, possibly due to long-run trade liberalization, industrial expansion, or structural shifts in sourcing.

2. Loss of Significance

- All other variables — including demand controls and policy dummies — become statistically insignificant once the time trend is included.
- This implies that their previously observed effects may have been confounded with time, and that the trend absorbs much of the explanatory power.

3. Model Fit

- R-squared improves to 0.362, up from 0.305 in the baseline model, confirming that the time trend adds explanatory value.
- However, the economic interpretation of other variables is weakened, as their coefficients are no longer statistically distinguishable from zero.

The inclusion of a linear time trend reveals that:

- Imports of barium chloride were rising over time, independent of short-term policy shocks or demand fluctuations.
- Once time is controlled for, none of the other variables remain statistically significant, suggesting that their effects may be non-robust or time-dependent.
- This highlights the importance of controlling for secular trends in time series analysis to avoid overstating the impact of short-term interventions.

(iii) In the equation estimated in part (ii), test for joint significance of all variables except the time trend. What do you conclude?

Hypothesis Tested

Performing an F-test on the following joint null hypothesis:

$$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = 0$$

This tests whether the six explanatory variables — $\log(\text{chempi})$, $\log(\text{gas})$, $\log(\text{rtwex})$, befile6 , affile6 , and afdec6 — are jointly insignificant, conditional on the presence of the time trend tt .

```
. test lchempi lgas lrtwex befile6 affile6 afdec6
```

```
( 1)  lchempi = 0
( 2)  lgas = 0
( 3)  lrtwex = 0
( 4)  befile6 = 0
( 5)  affile6 = 0
( 6)  afdec6 = 0
```

```
F( 6, 123) = 0.54
Prob > F = 0.7767
```

Econometric Interpretation

- The p-value of 0.7767 is far above conventional significance thresholds (0.10, 0.05, 0.01), meaning we fail to reject the null hypothesis.
- This implies that the six variables — demand-side controls and policy dummies — are jointly insignificant once the time trend is included.
- In other words, the time trend alone explains the variation in imports, and the remaining variables do not add meaningful explanatory power in this specification.

The joint significance test confirms that:

- None of the explanatory variables beyond the time trend are jointly significant.
- This reinforces the finding from Question (ii) that the time trend dominates the model.

- From a modeling perspective, this suggests that simplifying the specification to focus on temporal dynamics may be more appropriate than attributing variation to short-term demand or policy shocks.

(iv) The data is a monthly data that have not been seasonally adjusted. Therefore, we should add seasonal dummy variables to make sure none of the important conclusions change. It could be that the months just before the suit was filed are months where imports are higher or lower, on average, than in other months. Add monthly dummy variables to the equation in (iii) and test for seasonality (test their joint significance). Does including the monthly dummies change any other estimates or their standard errors in important ways?

```
. reg lchnimp lchempi lgas lrtwex befile6 affile6 afdec6 t feb mar apr may jun
jul aug sep oct nov dec
```

Source	SS	df	MS	Number of obs	=	131
-----+				F(18, 112)	=	4.33
Model	26.1336807	18	1.45187115	Prob > F	=	0.0000
Residual	37.5185675	112	.33498721	R-squared	=	0.4106
-----+				Adj R-squared	=	0.3158
Total	63.6522483	130	.489632679	Root MSE	=	.57878

lchnimp	Coefficient	Std. err.	t	P> t	[95% conf. interval]
-----+					
lchempi	-.4516555	1.271528	-0.36	0.723	-2.971026 2.067715
lgas	-.820624	1.345056	-0.61	0.543	-3.485679 1.844431
lrtwex	-.1971415	.5295314	-0.37	0.710	-1.24634 .852057
beffile6	.1648509	.2569789	0.64	0.523	-.3443198 .6740216
affile6	.1534004	.2719856	0.56	0.574	-.3855043 .692305
afdec6	-.2950163	.2994276	-0.99	0.327	-.8882937 .298261
t	.0123389	.0039163	3.15	0.002	.0045793 .0200985
feb	-.3554148	.293754	-1.21	0.229	-.9374507 .2266211
mar	.062566	.254858	0.25	0.807	-.4424025 .5675345
apr	-.4406149	.258398	-1.71	0.091	-.9525974 .0713676
may	.031299	.2591998	0.12	0.904	-.4822721 .5448702
jun	-.20095	.2592134	-0.78	0.440	-.7145481 .312648
jul	.0111115	.2683777	0.04	0.967	-.5206446 .5428675
aug	-.1271137	.2677917	-0.47	0.636	-.6577086 .4034812
sep	-.0751929	.2583502	-0.29	0.772	-.5870807 .4366949
oct	.0797627	.2570514	0.31	0.757	-.4295517 .5890771
nov	-.2603032	.2530623	-1.03	0.306	-.7617136 .2411073
dec	.0965326	.2615525	0.37	0.713	-.4217002 .6147654
_cons	27.30007	31.39707	0.87	0.386	-34.90919 89.50934

```
.
. test feb mar apr may jun jul aug sep oct nov dec
```

```
( 1)  feb = 0
( 2)  mar = 0
( 3)  apr = 0
( 4)  may = 0
( 5)  jun = 0
( 6)  jul = 0
( 7)  aug = 0
( 8)  sep = 0
( 9)  oct = 0
(10)  nov = 0
(11)  dec = 0
```

```
F( 11, 112) = 0.85
Prob > F = 0.5943
```

Econometric Interpretation

1. No Evidence of Seasonality

- The p-value of 0.5943 indicates that the monthly dummies are jointly insignificant.
- This means there is no systematic seasonal variation in Chinese barium chloride imports across months.
- Imports in months just before the antidumping filing (e.g., September–October 1983) are not statistically different from other months on average.

2. Stability of Other Estimates

- The time trend (t) remains statistically significant ($p = 0.002$), confirming its robustness.
- All other variables — including demand controls and policy dummies — remain statistically insignificant, consistent with earlier findings.
- The inclusion of monthly dummies does not materially alter the magnitude or standard errors of other coefficients.

3. Model Fit

- R-squared increases to 0.411, up from 0.362 in the previous model, but this improvement is modest and not driven by seasonal effects.
- The adjusted R-squared remains relatively low (0.316), reinforcing that the time trend is the dominant explanatory factor.

The seasonality test confirms that:

- Monthly variation does not explain import behavior, and there is no evidence of seasonal bias in the months surrounding the antidumping filing.

- Including monthly dummies does not change the significance or interpretation of other variables.
- The earlier conclusions — that the time trend dominates and policy/demand variables are not robustly significant — remain valid.

(v) In the equation estimated in part (ii), generate the residuals and also generate its one, two and three lags. Use these to construct the test for serial correlation for AR(1) and AR(3). Do you think the data have serial correlation problem?

```
. tsset t

Time variable: t, 1 to 131
      Delta: 1 unit

. predict u, res

. gen u_1 = L1.u
(1 missing value generated)

.
. gen u_2 = L2.u
(2 missing values generated)

.
. gen u_3 = L3.u
(3 missing values generated)

.
. reg u u_1 u_2 u_3
```

Source		SS	df	MS	Number of obs	=	128
-----+-----					F(3, 124)	=	3.63
Model		3.20416087	3	1.06805362	Prob > F	=	0.0149
Residual		36.5049251	124	.294394557	R-squared	=	0.0807
-----+-----					Adj R-squared	=	0.0584
Total		39.709086	127	.312669968	Root MSE	=	.54258

u		Coefficient	Std. err.	t	P> t	[95% conf. interval]
-----+-----						
u_1		.1787472	.0892215	2.00	0.047	.0021529 .3553415
u_2		.1126949	.0890726	1.27	0.208	-.0636047 .2889944
u_3		.1138384	.0882145	1.29	0.199	-.0607627 .2884396
_cons		.0073357	.0479617	0.15	0.879	-.0875939 .1022653
-----+-----						

Econometric Interpretation

1. Evidence of AR(1) Serial Correlation

- The coefficient on u_1 is statistically significant at the 5% level ($p = 0.047$), indicating that residuals exhibit first-order autocorrelation.
- This violates the classical OLS assumption of uncorrelated errors, which can lead to biased standard errors and invalid inference.

2. No Strong AR(2) or AR(3) Effects

- Coefficients on u_2 and u_3 are not statistically significant, suggesting that higher-order autocorrelation is not a major concern.

3. Joint Significance

- The F-test for joint significance of all three lags yields $p = 0.0149$, confirming that serial correlation is present overall, even if only AR(1) is individually significant.

The residual diagnostics reveal that:

- The model suffers from AR(1) serial correlation, which can distort standard errors and hypothesis tests.
- To correct for this, you should use robust standard errors, such as Newey-West HAC estimators, especially given the monthly frequency of the data.
- This finding justifies the use of *newey* in Question (vi), which we'll explore next.

(iv) Use the HAC standard errors from the command *newey* with 12 lags (since it is a monthly data) Does this change any other estimates or their standard errors in important ways?

```
newey lchnimp lchempi lgas lrtwex befile6 affile6 afdec6 t, lag(12)
```

Re-estimated the time-trend model using Newey–West HAC standard errors with 12 lags to correct for:

- Autocorrelation, as identified in Question (v).
- Heteroskedasticity, which may be present in monthly trade data.

This allows for more reliable inference, especially in the presence of serial correlation.


```
. newey lchnimp lchempi lgas lrtwex befile6 affile6 afdec6 t, lag(12)
```

Regression with Newey-West standard errors	Number of obs	=	131
Maximum lag = 12	F(7, 123)	=	10.15
	Prob > F	=	0.0000

		Newey-West				
	lchnimp	Coefficient	std. err.	t	P> t	[95% conf. interval]
lchempi		-.6862364	1.586569	-0.43	0.666	-3.826752 2.45428
lgas		.4656786	1.036104	0.45	0.654	-1.585225 2.516582
lrtwex		.0782237	.3008764	0.26	0.795	-.5173426 .6737901
beffile6		.09047	.100685	0.90	0.371	-.1088298 .2897697
affile6		.0970062	.1681771	0.58	0.565	-.2358902 .4299025
afdec6		-.3515018	.1744352	-2.02	0.046	-.6967856 -.006218
t		.0127058	.0048415	2.62	0.010	.0031225 .0222892
_cons		-2.367526	20.72287	-0.11	0.909	-43.38717 38.65212

Econometric Interpretation

1. Improved Inference for *afdec6*

- The post-decision dummy (*afdec6*) becomes statistically significant at the 5% level ($p = 0.046$) under HAC standard errors.
- This reinforces the conclusion that the ITC ruling led to a meaningful reduction in imports, even after correcting for autocorrelation.

2. Time Trend Robustness

- The time trend remains significant ($p = 0.010$), confirming a persistent upward drift in imports over time.
- This validates earlier findings and suggests that structural growth in imports is not an artifact of autocorrelated errors.

3. Other Variables Remain Insignificant

- Demand-side controls and pre/post-filing dummies remain statistically insignificant.
- Their standard errors increase slightly (for demand variables) or decrease (for dummies), but not enough to alter inference.

4. Model Fit and Stability

- The F-statistic remains strong (10.15, $p < 0.001$), indicating that the model is jointly significant.
- Coefficient estimates are stable, and HAC adjustment primarily affects standard errors.

Applying HAC standard errors confirms that:

- *afdec6* is now statistically significant, strengthening the evidence that the ITC decision reduced imports.
- The time trend remains robust, capturing long-run growth in import volumes.
- Other variables remain economically and statistically weak, and their insignificance is not due to autocorrelation.

This adjustment enhances the credibility of your findings and supports the use of robust inference in time series models.